Lab 01

Getting Started with WSO2 API Manager

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue.

Training Objective

Verify that the products required for these exercises are installed and configured, start WSO2 API Manager, and confirm that you can open its portals.

Note: The participants are expected to be connected to the internet throughout in order to successfully complete the lab exercises.

Business Scenario

PizzaShack Limited wants to extend their website for placing and managing online orders as a part of their effort in becoming the #1 online pizza shop. They have also found it increasingly useful to build an application for smartphones. The application is a web application allowing you to choose and buy a pizza online. They have subcontracted the development of the smartphone application to FunkyApps LLC. John Doe, Chief Architect of FunkyApps had some interesting feedback for PizzaShack. He suggested that the company considers monitoring of consumer statistics and probably looking into complex event processing in the future. John also suggested that they make use of an API Store backed by a modern API Gateway providing security features such as OAuth 2.0 access tokens.

In order to achieve this, PizzaShack will be using WSO2 API Manager and a number of other WSO2 products for monitoring statistics, single sign-on, and so on.

The application uses an API with three resources (menu, order and order details), exposed through API Manager. The PizzaShack back-end service ships with WSO2 API Manager as a sample, so no separate back end is needed. API analytics can optionally be sent to Moesif.

High-Level Steps

- Install WSO2 API Manager
- Configure API Manager Analytics (optional)
- Other installations
- Overview of the key directories in WSO2 API Manager

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- Key configuration files
- Configure port offsets
- Start API Manager and open the portals

Part 1 – Install WSO2 API Manager

Before installing, make sure each laptop meets the prerequisites below. Then extract the API Manager 4.7.0 zip file provided by the trainer (it runs on Linux, macOS and Windows).

1. Install a tested JDK: **Temurin OpenJDK 21 or 25**, and set `JAVA_HOME`.
2. Check the minimum resources for one API Manager instance: **4 GB RAM** (2 GB for the JVM) and **10 GB** of free disk space.
3. Extract the API Manager 4.7.0 zip file. The extracted folder is referred to as `<APIM_HOME>` in these labs.

Part 2 – Configure API Manager Analytics (Optional)

Analytics is not needed for the Day 1 labs. To try it, create a Moesif account and copy its **Application ID**, then add the following to `<APIM_HOME>/repository/conf/deployment.toml` and restart API Manager:

```
[apim.analytics]
enable = true
type = "moesif"

[apim.analytics.properties]
moesifKey = "<MOESIF_APPLICATION_ID>"
```

Part 3 – Other Installations

To complete the labs, install:

- A REST API client such as **Postman** or **cURL** (see below).
- The **API Controller (apictl)** command-line tool.

Database drivers are not needed: API Manager uses its built-in H2 database by default.

For macOS

cURL is already installed on macOS. To install or update it with Homebrew (install Homebrew from https://brew.sh/), run:

```
brew install curl
```

For Windows

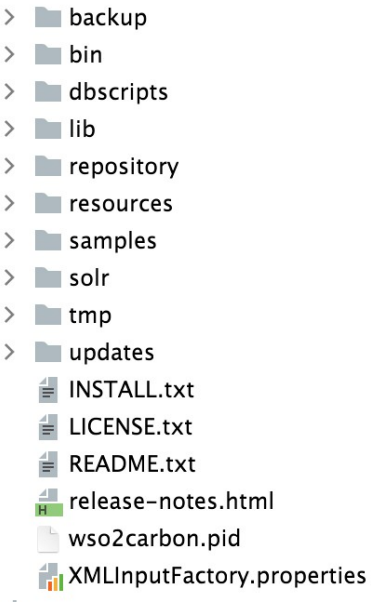
Windows 10 and later include `curl.exe`. On older versions, install cURL from the cURL website.

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Part 4 – Overview of the Key Directories in WSO2 API Manager

The structure of the `<APIM_HOME>` folder is as follows.

The <APIM_HOME> folder



- **`bin`** – All the executable files, including the scripts used to start and stop the server on Linux and Windows, for example api-manager.sh and api-manager.bat.
- **`dbscripts`** – A collection of database scripts required to create the Carbon database and the API Manager specific database on a variety of database management systems.
- **`lib`** – All the jar files that are converted to OSGi bundles at startup and copied to the dropins directory.
- **`repository`** – The main repository for all kinds of deployments and configurations in Carbon. This includes all default services, created APIs, Carbon configurations, etc.
- **`resources`** – Additional resources that may be used by API Manager.
- **`diagnostics-tool`** – A tool that collects diagnostic data (thread dumps, logs) when the server has problems.
- **`tmp`** – Temporary files that are created when the product runs. These files are cleared from time to time by housekeeping tasks.
- **`updates`** – Files used by the WSO2 Update tool to keep the product up to date.

Part 5 – Key Configuration Files

File	Description
`<APIM_HOME>/repository/conf/deployment.toml`	The main configuration file.
`<APIM_HOME>/repository/conf/log4j2.properties`	The log4j2 configuration file used by WSO2 Carbon.

Part 6 – Configure Port Offset

When you run multiple WSO2 products, multiple instances of the same product, or multiple WSO2 product clusters on the same server or virtual machines (VMs), you must change their default ports with an offset value to avoid port conflicts. The default HTTP and HTTPS ports (without offset) of a

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WSO2 product are 9763 and 9443, respectively. Port offset defines the number by which all ports defined in the runtime such as the HTTP/S ports will be changed. For example, if the default HTTP port is 9763 and the port offset is 1, the effective HTTP port will change to 9764. For each additional WSO2 product instance, you set the port offset to a unique value. The default port offset is 0.

There are two ways to set an offset to a port:

- Pass the port offset to the server during startup. The following command starts the server with all default ports incremented by 3:

  Linux / macOS

  ```
  sh api-manager.sh -DportOffset=3
  ```

  Windows

  ```
  api-manager.bat -DportOffset=3
  ```

- Set the `[server]` section of `<APIM_HOME>/repository/conf/deployment.toml` as follows:

  ```
  [server]
  offset = 3
  ```

With offset 3, the HTTPS port becomes 9446, the gateway HTTPS port 8246 and the gateway HTTP port 8283 (each port is its default plus the offset).

In these exercises, API Manager runs on each participant's laptop (or on the shared training server) with no offset. Analytics with Moesif is optional.

File	Port Offset
`<APIM_HOME>/repository/conf/deployment.toml`	0

Part 7 – Start API Manager and Open the Portals

4. Start WSO2 API Manager:

Linux / macOS

```
cd <APIM_HOME>/bin
sh api-manager.sh
```

Example – Linux / macOS:

```
cd ~/wso2am-4.7.0/bin
sh api-manager.sh
```

Windows

```
cd <APIM_HOME>\bin
api-manager.bat --run
```

Example – Windows:

```
cd C:\wso2am-4.7.0\bin
api-manager.bat --run
```

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5. Wait until the log shows `WSO2 Carbon started in` followed by the start-up time in seconds. The console then prints the portal URLs.

6. Open each URL below and sign in with the admin account (default user name and password: **admin** / **admin**):

Portal	URL
Publisher	`https://<APIM_HOST>:<HTTPS_PORT>/publisher`
Developer Portal	`https://<APIM_HOST>:<HTTPS_PORT>/devportal`
Admin Portal	`https://<APIM_HOST>:<HTTPS_PORT>/admin`
Management Console	`https://<APIM_HOST>:<HTTPS_PORT>/carbon`

7. Check the sample back end used in the next labs. This URL should return a JSON list of pizzas:

```
https://<APIM_HOST>:<HTTPS_PORT>/am/sample/pizzashack/v1/api/menu
```

Example:

```
https://localhost:9443/am/sample/pizzashack/v1/api/menu
```

8. To stop the server, press **Ctrl+C** in the terminal (or run `sh api-manager.sh stop` if you started it with `start`).

Expected Outcome

API Manager starts with no port offset, so the portals open on port 9443 and the gateway listens on 8243 (HTTPS) and 8280 (HTTP). You can sign in to the Publisher, Developer Portal, Admin Portal and Management Console, and the PizzaShack sample back end returns the menu.

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